Edge-Computing Video Analytics for Real-Time Traffic Monitoring:

A Case Study Analysis

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A. Introduction and Objectives

The Liverpool Smart Pedestrians project addressed critical urban planning challenges in Liverpool, NSW, Australia—a rapidly growing suburb expecting 30,000 additional daily pedestrians from CBD redevelopment. The project's primary objectives included developing a privacy-compliant edge-computing visual sensor using computer vision and deep neural networks to track multi-modal transportation (pedestrians, cyclists, vehicles) in real-time, leveraging existing CCTV infrastructure to reduce costs, and creating a scalable, interoperable data framework (Barthélemy et al., 2019).

Traditional traffic monitoring methods proved inadequate for capturing multi-modal patterns while respecting Australian privacy regulations limiting CCTV access to police and accredited operators. The project innovatively transformed underutilized surveillance infrastructure into valuable urban planning tools without compromising citizen privacy.

B. Methodology

The research employed a community-centered, requirements-driven approach beginning with two workshops (35 total participants) that established four core requirements: multi-modal detection, privacy compliance, infrastructure leverage, and scalability (Barthélemy et al., 2019).

Hardware development centered on the NVIDIA Jetson TX2 (15W power consumption, 256-core Pascal GPU) paired with a Pycom LoPy 4 module for LoRaWAN communications, packaged in waterproof IP67-rated cases for outdoor deployment. The software architecture implemented a four-step iterative process at approximately 20 frames per second: frame acquisition, object detection using YOLO V3, tracking via SORT algorithm, and database updating with periodic transmission via Ethernet or LoRaWAN.

C. Technology and Implementation

Hardware Selection

The NVIDIA Jetson TX2 provided optimal balance between computational power and energy efficiency, featuring quad-core Cortex-A57 CPU, dual-core Denver2 CPU, and 256-core Pascal GPU with 8GB memory—essential for real-time neural network inference (Barthélemy et al., 2019). The specialized CUDA cores accelerated deep learning computations critical for YOLO V3 processing. The Pycom LoPy 4's LoRaWAN capability enabled deployment beyond fixed network infrastructure, while dual connectivity (Ethernet + LoRaWAN) provided flexibility and redundancy.

Software Architecture

YOLO V3 (You Only Look Once version 3) was selected for object detection due to its optimal speed-accuracy balance, performing detection in a single forward pass through a 106-layer Darknet network (Redmon & Farhadi, 2018). The algorithm divides 416×416 pixel input images into three detection grids (13×13, 26×26, 52×52 cells) enabling multi-scale detection—crucial for identifying objects at varying distances. Each grid cell predicts three bounding boxes with coordinates, confidence scores, and class probabilities for six object types: pedestrians, bicycles, cars, buses, trucks, and motorbikes.

The SORT (Simple Online and Real-time Tracking) algorithm maintains object identity across frames using Kalman filters for motion prediction and the Hungarian algorithm for optimal detection-to-track assignment (Bewley et al., 2016). This combination enables trajectory extraction while maintaining computational efficiency.

Edge-Computing Advantages

The edge-computing paradigm provides critical benefits aligned with smart city principles (Satyanarayanan et al., 2015; Shi et al., 2016):

1. **Privacy preservation:** On-device processing ensures only metadata (counts, trajectories, timestamps) transmits to the cloud
2. **Bandwidth efficiency:** Minimal outbound traffic reduces network infrastructure requirements
3. **Latency reduction:** Real-time local processing enables immediate decision-making
4. **Scalability:** Distributed processing prevents centralized bottlenecks

D. Validation and Performance

Accuracy Assessment

Validation using the Oxford Town Center Dataset (1920×1080 @ 25fps, 230 annotated pedestrians over 4,500 frames) revealed 69% mean accuracy with 33% median relative error (Barthélemy et al., 2019). The system consistently underestimated counts (mean error: -5.34 pedestrians per frame), a conservative bias acceptable for trend detection applications. Accuracy correlated inversely with crowd density: exceeding 80% for small groups (6-10 people) but dropping to 40-60% in dense crowds (20-28 people) due to occlusion-triggered Non-Maximum Suppression.

Processing Performance

The system maintained 19.57 fps mean processing rate with peaks at 22.99 fps. Anti-correlation between detection count and speed revealed the CPU-based SORT algorithm as a bottleneck—fewer detections (2-8 objects) enabled 22-23 fps while dense crowds (18-20 objects) reduced performance to 15-17 fps. GPU utilization fluctuated between 40-100%, with drops corresponding to CPU bottlenecks during intensive tracking operations.

E. Real-World Applications

Indoor Emergency Evacuation Monitoring

Indoor deployment at the SMART Infrastructure Facility during a fire alarm event (14:30-15:30) detected 631 unique individuals and provided critical insights: zero detections during the alarm confirmed complete evacuation, and the second peak showed 25-30% fewer people than initial evacuation, indicating partial non-return (Barthélemy et al., 2019). Trajectory visualization confirmed logical movement patterns, while heat mapping identified high-density zones valuable for capacity planning and emergency exit design.

Outdoor Urban Traffic Monitoring

Liverpool's 20-sensor deployment (15 existing CCTVs, 5 mobile units) provided comprehensive traffic monitoring. Analysis of one intersection over eight days revealed clear circadian rhythms with pedestrian peaks around noon (60-70 detections/hour) and vehicle bimodal peaks at 09:00 and 16:00. Sunday showed lowest pedestrian activity, confirming expected patterns. On the busiest day (February 23, 2019), 20,399 unique objects were detected. Coordinate plotting revealed two distinct pedestrian flows, confirming crosswalk usage while identifying off-crosswalk behavior. Minimal bicycle detection validated community feedback about limited cycling in the area.

These deployments demonstrate the integration of sensor technologies and edge gateways by enabling infrastructure optimization through existing CCTV networks, multi-sensor coordination via the OM2M-based Agnosticity framework, real-time decision support dashboards for urban planners, and co-located sensing with air quality and noise sensors for correlation analysis.

F. Challenges and Future Work

Project Challenges

Key challenges included accuracy degradation in dense crowds due to YOLO V3's Non-Maximum Suppression, CPU-based SORT implementation creating processing bottlenecks, inability to re-identify objects across camera transitions, and consistent detection underestimation requiring calibration (Barthélemy et al., 2019). The authors proposed GPU-accelerated tracking, migration to production frameworks (TensorFlow with TensorRT), hardware upgrades to NVIDIA Jetson Xavier, error quantification models, and advanced MOT algorithms.

Post-2019 Technological Developments

Significant advancements since 2019 could substantially enhance system performance:

AI Algorithms: YOLOv8 (2023) offers 20-30% accuracy improvements; transformer-based detection (DETR) eliminates hand-crafted components like NMS; efficient networks optimize the accuracy-speed-size tradeoff for edge deployment.

Edge Hardware: NVIDIA Jetson Orin (2022) provides up to 275 TOPS compared to TX2's 1.33 TFLOPS, enabling 100+ fps; specialized AI accelerators (Google Coral, Intel Movidius) provide dedicated neural network processing at lower power; neuromorphic processors enable ultra-low-latency, energy-efficient inference.

Communication Protocols: 5G integration enables hybrid edge-cloud architectures; LoRaWAN enhancements provide better latency and synchronization; MQTT 5.0 improves IoT messaging scalability.

Privacy Technologies: Federated learning enables collaborative training without sharing raw data; differential privacy quantifies and limits privacy risks; advanced on-device anonymization completely eliminates identifiable features.

Computer Vision: ByteTrack, BoT-SORT, and StrongSORT achieve 75-80% tracking accuracy; monocular depth estimation enables more accurate counting in crowds; self-supervised anomaly detection identifies unusual patterns without explicit training.

A 2026 implementation could achieve 90%+ accuracy even in dense crowds, 60+ fps processing, sub-100ms latency for real-time applications, and enhanced privacy through hardware-enforced trusted execution environments.

G. Personal Evaluation

Project Success

The Liverpool Smart Pedestrians project successfully translated academic research into practical urban planning tools through community-centered design addressing actual stakeholder needs, pragmatic innovation leveraging existing infrastructure, privacy-by-design enabling compliant deployment, and open-source approaches enhancing reproducibility and adoption (Barthélemy et al., 2019). The deployment provided urban planners with previously unavailable real-time, multi-modal traffic data, representing substantial improvement over periodic manual counts or vehicle-only sensors.

Critical Assessment

Despite successes, the 69% mean accuracy falls short for applications demanding precise counts, and 40-60% accuracy in dense crowds limits applicability in high-traffic scenarios. The inability to track individuals across cameras prevents network-wide journey analysis. The case study lacks weather resilience assessment—critical for outdoor deployments. Computational constraints suggest architectural inefficiencies addressable with modern unified memory architectures.

Recommended Enhancements

Building on the authors' suggestions, I recommend: (1) adaptive confidence thresholding based on detected crowd density to capture more true positives in dense scenarios, (2) appearance-based re-identification networks enabling city-wide movement pattern analysis, (3) active learning pipelines where low-confidence detections undergo human review for continuous improvement, (4) weather-aware models or domain adaptation maintaining performance across environmental variations, (5) time-series forecasting (LSTM, Transformer models) predicting traffic 15-30 minutes ahead for proactive management, (6) dynamic frame rate adjustment based on detected activity for energy optimization, and (7) behavioral analytics extracting insights like pedestrian wait times and near-miss detection informing safety interventions.

Broader Implications

This project exemplifies smart city best practices: incremental implementation starting with well-defined use cases, stakeholder engagement building trust, interoperability focus preventing vendor lock-in, and privacy prioritization demonstrating that valuable insights can be extracted without compromising citizen rights. The open-source approach and moderate hardware costs ($500-700 per sensor) make replication feasible for municipalities of various sizes, though successful adoption requires local calibration, regulatory compliance assessment, technical capacity development, and sustained political commitment beyond pilot phases.

Conclusion

The Liverpool Smart Pedestrians project successfully demonstrates that edge-computing video analytics can provide valuable traffic monitoring while respecting privacy and leveraging existing infrastructure. The 69% accuracy, while imperfect, proves sufficient for identifying trends informing urban planning decisions. With substantial technological advancements since 2019, municipalities implementing similar systems today could achieve significantly improved performance while maintaining the core architectural principles that made Liverpool's implementation successful. The project underscores that smart city success requires balancing technological capability with practical constraints while focusing on solving specific, well-defined problems rather than pursuing technology for its own sake.

References

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